

I.N.A.Y.A.T. Project Deep Dive & Status Report

This official status report provides a complete deep dive audit of the MCA 4th Semester Final Project titled I.N.A.Y.A.T. (Intelligent Neural Architecture for Yielding Agentic Thinking). No project code was changed in this step. This document outlines completed work, remaining tasks, critical questions raised, and missing elements to guide the next phases of development.

1. WORK COMPLETED (WHAT HAS BEEN DONE)

- **Full Project Scaffolding:** Created the directory layout and marked all 21 key files (app.py, core modules, tests, warmup script, docs, etc.) with production-grade imports, docstrings, and structures.
- **Windows + Python 3.14 Compatibility:** Resolved Python 3.14 compilation crashes on Windows by locking and building compatible packages in the virtual environment. Fixed compilation errors in numpy (2.4.6) and pillow (12.2.0) by using pre-built wheels.
- **Migrated Gemini Integration:** Migrated LlamaIndex from deprecated llama-index-llms-gemini and llama-index-embeddings-gemini integrations to the modern llama-index-llms-google-genai and llama-index-embeddings-google-genai integrations, resolving compilation Pillow version blockers.
- **Corrected Mem0 Client Parameters:** Fixed core/memory.py. Replaced deprecated top-level user_id arguments with the required query filters={'user_id': user_id} parameter, avoiding immediate client crashes.
- **Fixed Mem0 Client Result Parsing:** Fixed a critical bug where dict keys were iterated instead of results. The client now correctly extracts consolidated memories from the SDK v2 paginated dictionary format using raw.get('results', []).
- **Circuit Breaker Sentinel Pattern:** Updated core/memory.py and core/graph_store.py to use a sentinel pattern (sentinel = object()) so that empty queries or empty memory histories do not count as failures or trip the circuit breaker.
- **Resolved Logging Traceback Glitch:** Replaced the right-arrow Unicode character (u2192) in warmup.py with ASCII arrows (->), resolving terminal UnicodeEncodeErrors on legacy Windows encodings.
- **Smoke Test Execution:** Successfully executed the 17 smoke tests in tests/smoke_test.py inside the virtual environment. All tests passed. Staged and committed changes in git.

2. WORK NOT DONE (WHAT IS LEFT TO DO)

- **Active Environment Keys (.env):** No actual credentials have been added yet. The dummy .env was deleted to preserve startup gate testing. The user must copy .env.example to .env and insert their real GEMINI_API_KEY, MEM0_API_KEY, and Neo4j login credentials.
- **Upload Documents:** The data/documents/ directory is currently empty. To use the PropertyGraphIndex and RAG query engine, user PDF files must be uploaded to this directory.
- **Neo4j Instance Setup:** The Neo4j database instance is not yet populated or initialized with user nodes or relationships. Connecting with real credentials will trigger LlamaIndex auto-indexing on boot.
- **Verify Graph Visualizations:** Visual check of the knowledge graph inside the Neo4j Aura console or streamlit display represents a manual step that is pending real database connection.

3. AI OPEN QUESTIONS & DECISIONS REQUIRED

- **LLM and Embedding Models Selection:** Currently configured to use GoogleGenAI('gemini-1.5-flash') and GoogleGenAIEmbedding('text-embedding-004'). Do you prefer using other versions such as 'gemini-3.5-flash' or 'gemini-embedding-2-preview'?
- **UI Reload Button Behavior:** The reload system button in the sidebar reload mechanism works. Do you want to add further system resets (e.g. purging Neo4j DB besides clearing Mem0 memories)?
- **Document Indexing Strategy:** Currently, core/agent.py automatically indexes files in data/documents/ at boot. Do you want to index files dynamically on upload inside the Streamlit page instead?

4. QUESTIONS YOU HAVE TO ASK (ADVISOR/EXAMINER PREP)

- **Graph Visualizer Format:** Ask: 'Should we showcase the Neo4j knowledge graph visualization directly inside the Streamlit UI (using PyVis or an iframe to Neo4j Bloom/Console), or is showcasing the query-response and matching nodes textually sufficient for the MCA final grading?'
- **Deployment Strategy:** Ask: 'Does the final presentation require cloud deployment (e.g., Streamlit Community Cloud / HuggingFace Spaces), or is local execution on our laptops acceptable since Neo4j and Mem0 are hosted in the cloud?'
- **Document Indexing Actions:** Ask: 'Should the agent support document deletion or incremental indexing, or is simple boot-time indexing of the data/documents/ folder sufficient?'

5. WHAT IS MISSING (GAP ANALYSIS)

- **Interactive Graph Canvas in UI:** The system queries Neo4j, but there is no graphical visualization (like a 2D canvas of nodes and links) embedded directly in the Streamlit page. If required, a library like streamlit-agraph or PyVis needs to be integrated.
- **User Session Persistence:** The user ID is stored in the sidebar text input, but it resets if the app container restarts or if the page is hard reloaded. Cookie persistence would be required for a production-grade system.